列聯表分析與邏輯斯迴歸

Contingency Table Analysis and Logistic Regression

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1 Contingency table analysis

1.1 Pneumoconiosis dataset

Pneumoconiosis (塵肺病) is an occupational lung disease and a restrictive lung disease caused by the inhalation of dust, often in mines and from agriculture.

- `year`: Number of years of exposure
- `y`: 1 = severe case; 0 = otherwise

```
pneu <- read.csv("data/pneumoconiosis.csv")
pneu <- transform(pneu, yeargp = ifelse(year < 30, "<30", ">30"))
head(pneu)
```

```
  id year y yeargp
1  1 15.0 0    <30
2  2  5.8 0    <30
3  3 21.5 0    <30
4  4 51.5 0    >30
5  5  5.8 0    <30
6  6 46.0 1    >30
```

```
dim(pneu)
```

```
[1] 371   4
```

1.2 Contingency table object in R

```
pneu_tab <- xtabs(~ yeargp + y, data = pneu) # table(pneu$yeargp, pneu$y)
pneu_tab
```

```
      y
yeargp   0   1
   <30 231  12
   >30  96  32
```

```
class(pneu_tab)
```

```
[1] "xtabs" "table"
```

```
dim(pneu_tab)
```

```
[1] 2 2
```

1.3 Hypothesis test

- **Statement 1**
  - H_0 : 礦工暴露年數與得到塵肺病與否無關
  - H_1 : 礦工暴露年數與得到塵肺病與否有關
- **Statement 2**: 令 p_1 與 p_2 分別代表暴露年數 <30 年與 >30 年的礦工，得到塵肺病的機率
  - $H_0 : p_1 = p_2$
  - $H_1 : p_1 \neq p_2$
- **Statement 3**: 令 θ 為暴露年數 >30 年的礦工相較於暴露 <30 年的礦工，得到塵肺病的勝算比 (odds ratio, OR)
  - $H_0 : \theta = 1$
  - $H_1 : \theta \neq 1$

1.4 Pearson's chi-squared test

當利用 Pearson 近似卡方統計量對自由度為 1 的列聯表進行 獨立性 (independence)、同質性 (homogeneity)、或適合度 (goodness-of-fit) 檢定時，由於列聯表屬於計數型離散資料，而卡方分布為連續型機率分布，此時可採用 葉茲連續校正 (Yates's continuity correction) 來降低連續分布近似離散數據所產生的誤差。

```
chisq.test(pneu_tab) # default: correct = TRUE
```

```
    Pearson's Chi-squared test with Yates' continuity correction

data:  pneu_tab
X-squared = 30.389, df = 1, p-value = 3.536e-08
```

```
chisq.test(pneu_tab, correct = FALSE)
```

```
    Pearson's Chi-squared test

data:  pneu_tab
X-squared = 32.279, df = 1, p-value = 1.335e-08
```

Note

當所有細格內的期望次數 ≥ 10 時，連續校正對檢定結果的影響很小。

```
chisq.test(pneu_tab)$expected
```

```
      y
yeargp        0        1
   <30 214.1806 28.81941
   >30 112.8194 15.18059
```

1.5 Fisher's exact test

- 對於任意維度的列聯表，當有「80% 以上細格的期望次數 ≥ 5」時，卡方檢定統計量才會較好地近似卡方分布。若是 2×2 列聯表，即 $4 \times 0.8 = 3.2$ 個以上的細格 (也就是所有細格) 的期望次數都要 ≥ 5。若此原則不成立，建議改用 Fisher's exact test。
- Fisher's exact test 利用超幾何分布 (hypergeometric distribution) 計算列聯表中觀測次數的機率。超幾何分布屬於離散型機率分布，更適合描述計數資料。

Arguments of `fisher.test()`

- `alternative`: alternative hypothesis
  - `"two.sided"` ($H_0 : OR = 1$; $H_1 : OR \neq 1$)
  - `"greater"` ($H_0 : OR \leq 1$; $H_1 : OR > 1$)
  - `"less"` ($H_0 : OR \geq 1$; $H_1 : OR < 1$)
- `or = 1`: the hypothesized odds ratio
- `conf.int = TRUE`: whether a confidence interval for the odds ratio should be computed and returned
- `conf.level = 0.95`: confidence level

See `?fisher.test` for more details.

Two-sided test

```
fisher.test(pneu_tab)
```

```
    Fisher's Exact Test for Count Data

data:  pneu_tab
p-value = 4.088e-08
alternative hypothesis: true odds ratio is not equal to 1
95 percent confidence interval:
  3.046001 14.210736
sample estimates:
odds ratio
  6.380454
```

Right-tailed test

```
fisher.test(pneu_tab, alternative = "greater")
```

```
    Fisher's Exact Test for Count Data

data:  pneu_tab
p-value = 3.904e-08
alternative hypothesis: true odds ratio is greater than 1
95 percent confidence interval:
 3.383273      Inf
sample estimates:
odds ratio
  6.380454
```

Fisher's exact test 假設在列總和 (row margin) 與行總和 (column margin) 固定的情況下，2×2 列聯表的任一細格發生次數服從超幾何分布，其他細格的次數由邊際總和自動決定。

```
addmargins(pneu_tab)
```

year	Y=0	Y=1	Sum
<30	231	12	243
>30	96	**32**	128
Sum	327	44	371

在列總和 (243, 128) 與行總和 (327, 44) 固定，且虛無假設 ($H_0 : OR = 1$; 暴露時間不影響塵肺病) 成立的情況下，假設暴露大於 30 年的 case 人數 X (觀測值 32) 服從超幾何分布，則其機率質量函數為:

$$f(x) = P(X = x) = \frac{\binom{44}{x}\binom{327}{128-x}}{\binom{371}{128}}, \quad x = 0, 1, ..., 44$$

其直觀的解釋為: 若母體中包含 44 位 case 和 327 位 noncase，從中以「取後不放回」的方式隨機抽出 128 位樣本，則裡面包含 x 位 case 的機率。

1.5.1 Exact *p*-value

- Exact p-value for two-sided test ($H_1 : OR \neq 1$)

  > Two-sided tests are based on the probabilities of the tables, and take as "more extreme" all tables with probabilities less than or equal to that of the observed table, the p-value being the sum of such probabilities.
  >
  > From `?fisher.test`

  虛無假設 ($H_0 : OR = 1$) 成立下，得到「與目前的觀測值 ($X = 32$) 一樣」或「更極端」觀測值的機率。

$$p\text{-value} = \sum_{x: f(x) \leq f(32)} P(X = x)$$

```
p_all <- dhyper(0:44, 44, 327, 128)
p_obs <- dhyper(32, 44, 327, 128)

# Which values are the same as or more extreme than 32 ?
(0:44)[p_all <= p_obs]
```

```
 [1]  0 32 33 34 35 36 37 38 39 40 41 42 43 44
```

```
# p-value
sum(p_all[p_all <= p_obs])
```

```
[1] 4.088232e-08
```

- Exact p-value for right-tailed test ($H_1 : OR > 1$)

$$p_{\text{upper}} = \sum_{x=32}^{44} P(X = x)$$

```
sum(dhyper(32:44, 44, 327, 128))
```

```
[1] 3.903773e-08
```

- Exact p-value for left-tailed test ($H_1 : OR < 1$)

$$p_{\text{lower}} = \sum_{x=0}^{32} P(X = x)$$

```
sum(dhyper(0:32, 44, 327, 128))
```

```
[1] 1
```

1.5.2 Conditional maximum likelihood estimation (CMLE)

- Noncentral hypergeometric distribution

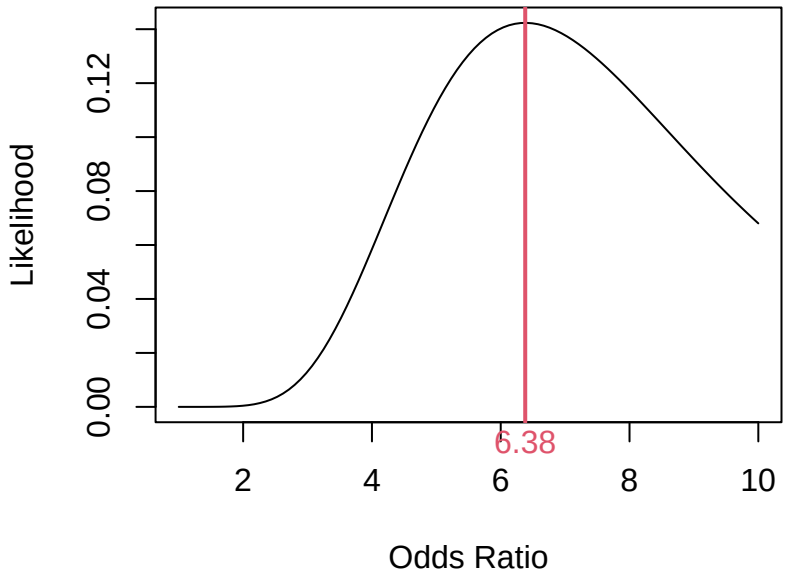
$$P(X = x) = \frac{\binom{44}{x}\binom{327}{128-x}OR^x}{\sum_{y=0}^{44}\binom{44}{y}\binom{327}{128-y}OR^y}, \quad x = 0, 1, ..., 44$$

```
pneu_pmf <- function(x, or) {
  x_all <- 0:44
  denom <- sum(choose(44, x_all) * choose(327, 128 - x_all) * or^x_all)
  L <- (choose(44, x) * choose(327, 128 - x) * or^x) / denom
  return(L)
}

or_cmle <- optimize(\(or) pneu_pmf(x = 32, or = or),
                    interval = c(1e-5, 1e2), maximum = TRUE)$maximum
or_cmle
```

```
[1] 6.380425
```

```r
plot(Vectorize(\(or) pneu_pmf(x = 32, or = or)), from = 1, to = 10,
     xlab = "Odds Ratio", ylab = "Likelihood")
abline(v = or_cmle, col = 2, lwd = 2)
mtext(round(or_cmle, 2), side = 1, at = or_cmle, col = 2)
```



1.5.3 Exact confidence interval

- Lower bound

```r
uniroot(\(or) sum(pneu_pmf(x = 32:44, or = or)) - 0.025,
        interval = c(1e-5, 1e2))$root
```

```
[1] 3.046096
```

- Upper bound

```r
uniroot(\(or) sum(pneu_pmf(x = 0:32, or = or)) - 0.025,
        interval = c(1e-5, 1e2))$root
```

```
[1] 14.20996
```

2 Logistic regression

類別型反應變數的記錄方式分為:

- **individual data**: 對每一個觀測對象記錄其解釋變數數值以及事件發生與否 (0/1)。
- **grouped data**: 將解釋變數數值相同的對象歸為一組，記錄該組的事件發生次數。

2.1 Individual data

前面使用的塵肺病資料 (pneu) 屬於 individual data。

```
head(pneu)
```

```
  id year y yeargp
1  1 15.0 0    <30
2  2  5.8 0    <30
3  3 21.5 0    <30
4  4 51.5 0    >30
5  5  5.8 0    <30
6  6 46.0 1    >30
```

2.1.1 Binary covariate

```
m1 <- glm(y ~ yeargp, family = binomial, data = pneu)
```

可用 summary() 檢視模型的詳細資訊：

```
summary(m1)
```

```
Call:
glm(formula = y ~ yeargp, family = binomial, data = pneu)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  -2.9575     0.2961  -9.990  < 2e-16 ***
yeargp>30     1.8589     0.3596   5.169 2.35e-07 ***
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 270.18  on 370  degrees of freedom
Residual deviance: 239.55  on 369  degrees of freedom
AIC: 243.55

Number of Fisher Scoring iterations: 5
```

Interpretation of **intercept** term

暴露年數 <30 年的礦工「得到塵肺病的機率」是「未得病機率」的 $e^{-2.9575} = 0.052$ 倍，即暴露年數 <30 年的礦工得到塵肺病的「勝算」(odds) 為 0.052。

- p-value $< 2 \times 10^{-16} < 0.05$
- 95% 信賴區間未包含 1

$$(e^{-2.9575-1.96\times0.2961}, e^{-2.9575+1.96\times0.2961}) = (0.029, 0.093)$$

檢定結果可以拒絕虛無假說，表示此「勝算」與 1 有顯著差異。

Interpretation of **slope** term

暴露年數 >30 年的礦工得到塵肺病的「勝算」(odds) 是暴露年數 <30 年的礦工的 $e^{1.8589} = 6.417$ 倍，即暴露年數 >30 年的礦工相較於暴露 <30 年的礦工，得到塵肺病的「勝算比」(odds ratio, OR) 為 6.417。

- p-value $= 2.35 \times 10^{-7} < 0.05$
- 95% 信賴區間未包含 1

$$(e^{1.8589-1.96\times0.3596}, e^{1.8589+1.96\times0.3596}) = (3.171, 12.984)$$

檢定結果可以拒絕虛無假說，表示此「勝算比」與 1 有顯著差異。

```
exp(coef(m1))
```

```
(Intercept)   yeargp>30
 0.05194805  6.41666662
```

```r
exp(confint.default(m1))
```

```
                2.5 %      97.5 %
(Intercept) 0.02907779  0.09280624
yeargp>30   3.17109526 12.98403459
```

> **Note**
>
> Logistic regression 的斜率係數即代表 $\log(OR)$。當解釋變數只有一個且是類別變數時，其係數估計值與標準誤 (standard error) 會與直接從列聯表計算的結果完全一致:
>
> $$OR = \frac{231 \times 32}{12 \times 96} = 6.417$$
> $$\log(OR) = \log(6.417) = 1.8589$$
> $$SE(\log(OR)) = \sqrt{\frac{1}{231} + \frac{1}{12} + \frac{1}{96} + \frac{1}{32}} = 0.3596$$

2.1.2 Continuous covariate

```r
m2 <- glm(y ~ year, family = binomial, data = pneu)
summary(m2)
```

```
Call:
glm(formula = y ~ year, family = binomial, data = pneu)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept) -4.79648    0.56859  -8.436  < 2e-16 ***
year         0.09346    0.01543   6.059 1.37e-09 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 270.18  on 370  degrees of freedom
Residual deviance: 219.33  on 369  degrees of freedom
```

```
AIC: 223.33

Number of Fisher Scoring iterations: 6
```

Interpretation of **slope** term

暴露年數每增加一年，得到塵肺病的「勝算比」(odds ratio, OR) 為 $e^{0.0935} = 1.098$。

- p-value $= 1.37 \times 10^{-9} < 0.05$
- 95% 信賴區間未包含 1

$$(e^{0.0935-1.96\times0.0154}, e^{0.0935+1.96\times0.0154}) = (1.065, 1.132)$$

檢定結果可以拒絕虛無假說，表示 OR 與 1 有顯著差異。

```r
exp(coef(m2))[-1]
```

```
   year
1.09797
```

```r
exp(confint.default(m2))[-1, ]
```

```
   2.5 %   97.5 %
1.065270 1.131674
```

Question

一位暴露年數為 20 年的礦工相較於暴露 5 年的礦工，得到塵肺病的勝算比是多少?

$$OR = \frac{odds|_{\text{year}=20}}{odds|_{\text{year}=5}}$$

$$\log(odds|_{\text{year}=20}) = -4.7965 + 0.0935 \times 20 \quad ...(1)$$

$$\log(odds|_{\text{year}=5}) = -4.7965 + 0.0935 \times 5 \quad ...(2)$$

$$(1) - (2) : \log(\frac{odds|_{\text{year}=20}}{odds|_{\text{year}=5}}) = 0.0935 \times (20 - 5) = 1.4025 = \log(OR)$$

$$OR = e^{1.4025} = 4.065$$

2.2 Grouped data

```
pneu_gp <- aggregate(cbind(case = y, noncase = 1-y) ~ year + yeargp,
                     data = pneu, FUN = sum)
pneu_gp$n <- pneu_gp$case + pneu_gp$noncase

pneu_gp
```

```
  year yeargp case noncase  n
1  5.8    <30    0      98 98
2 15.0    <30    1      53 54
3 21.5    <30    3      40 43
4 27.5    <30    8      40 48
5 33.5    >30    9      42 51
6 39.5    >30    8      30 38
7 46.0    >30   10      18 28
8 51.5    >30    5       6 11
```

```
pneu_gp[pneu_gp$year == 51.5, ]
```

```
  year yeargp case noncase  n
8 51.5    >30    5       6 11
```

```
pneu[pneu$year == 51.5, ]
```

```
     id year y yeargp
4     4 51.5 0    >30
31   31 51.5 0    >30
49   49 51.5 1    >30
51   51 51.5 0    >30
55   55 51.5 0    >30
93   93 51.5 0    >30
109 109 51.5 0    >30
134 134 51.5 1    >30
174 174 51.5 1    >30
323 323 51.5 1    >30
362 362 51.5 1    >30
```

對 grouped data 配適 logistic regression 有兩種語法:

1. 將模型的 response 設為一個 2-column matrix，分別代表 case 與 non-case 的人數

```
m3 <- glm(cbind(case, noncase) ~ yeargp, family = binomial, data = pneu_gp)
```

2. 將模型的 response 設為 case 的比例，但注意要用總人數加權

```
m3 <- glm(case/n ~ yeargp, family = binomial, data = pneu_gp, weights = n)
```

兩種語法完全等價。

> **Note**
>
> 對於相同的資料集，不論採用 grouped data 或 individual data 格式，所配適的模型係數與統計推論會完全一致。

- Binary covariate

```
m3 <- glm(case/n ~ yeargp, family = binomial, data = pneu_gp, weights = n)
coef(summary(m3))
```

```
             Estimate Std. Error   z value     Pr(>|z|)
(Intercept) -2.957511  0.2960740 -9.989093 1.701305e-23
yeargp>30    1.858899  0.3596199  5.169065 2.352681e-07
```

```
all.equal(coef(summary(m1)), coef(summary(m3)), tolerance = 1e-4)
```

```
[1] TRUE
```

- Continuous covariate

```
m4 <- glm(case/n ~ year, family = binomial, data = pneu_gp, weights = n)
coef(summary(m4))
```

```
               Estimate Std. Error   z value     Pr(>|z|)
(Intercept) -4.79648027 0.56858732 -8.435785 3.289876e-17
year         0.09346291 0.01542596  6.058808 1.371341e-09
```

```
all.equal(coef(summary(m2)), coef(summary(m4)), tolerance = 1e-4)
```

```
[1] TRUE
```

2.3 Prediction and confidence interval

Arguments of `predict()`

- `newdata`: a `data.frame` in which to look for variables with which to predict
- `type`:
  - `"link"` log-odds (probabilities on logit scale)
  - `"response"` probabilities
- `se.fit`: whether standard errors are returned
  - If `FALSE`, return a vector or matrix of predictions.
  - If `TRUE`, return a `list` with components
    - `$fit`: predictions
    - `$se.fit`: estimated standard errors

See `?predict.glm` for more details.

- Create a new dataset for predictions

```
pneu_new <- data.frame(year = seq(0, 100, by = 1))
```

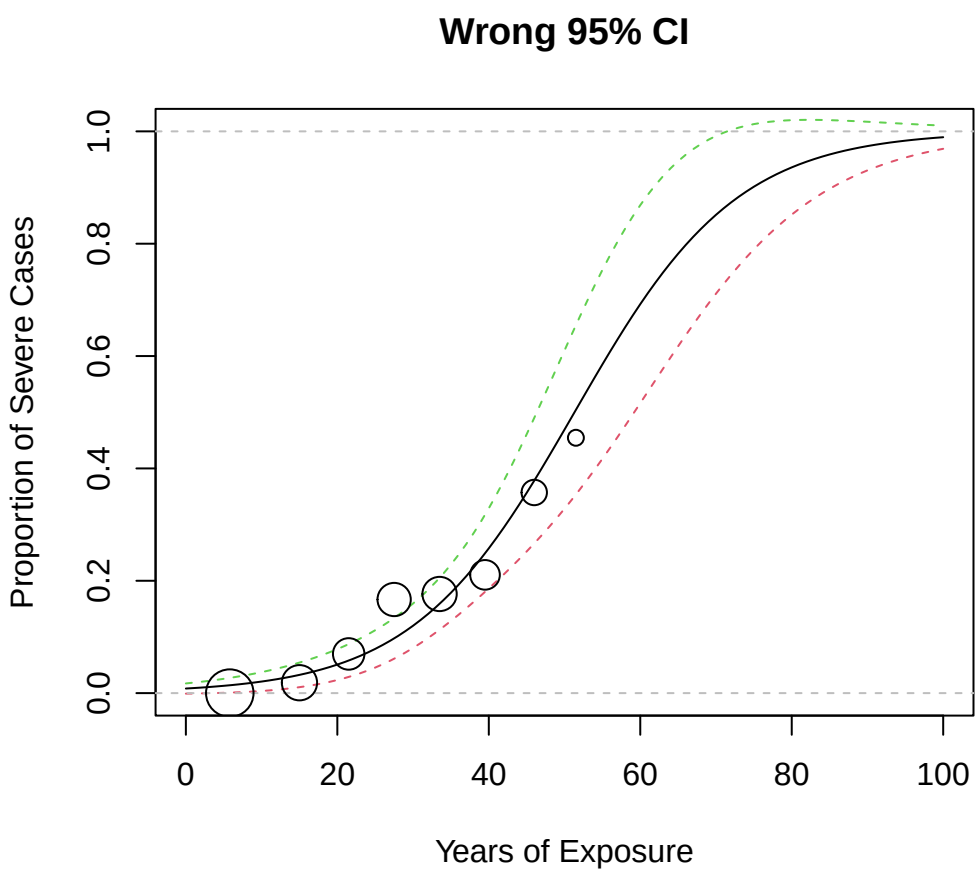
- Predict probabilities at different levels of `year` with confidence intervals

Wrong 95% CI

```
pred <- predict(m4, newdata = pneu_new,
                type = "response", se.fit = TRUE)

pneu_ci <- cbind(fit = pred$fit,
                 lower = pred$fit - 1.96 * pred$se.fit,
                 upper = pred$fit + 1.96 * pred$se.fit)

matplot(pneu_new$year, pneu_ci, type = 'l', lty = c(1, 2, 2), ylim = c(0, 1),
        xlab = "Years of Exposure", ylab = "Proportion of Severe Cases",
        main = "Wrong 95% CI")
abline(h = 0:1, lty = 2, col = "gray")
points(case/n ~ year, pneu_gp, cex = sqrt(n)/3)
```

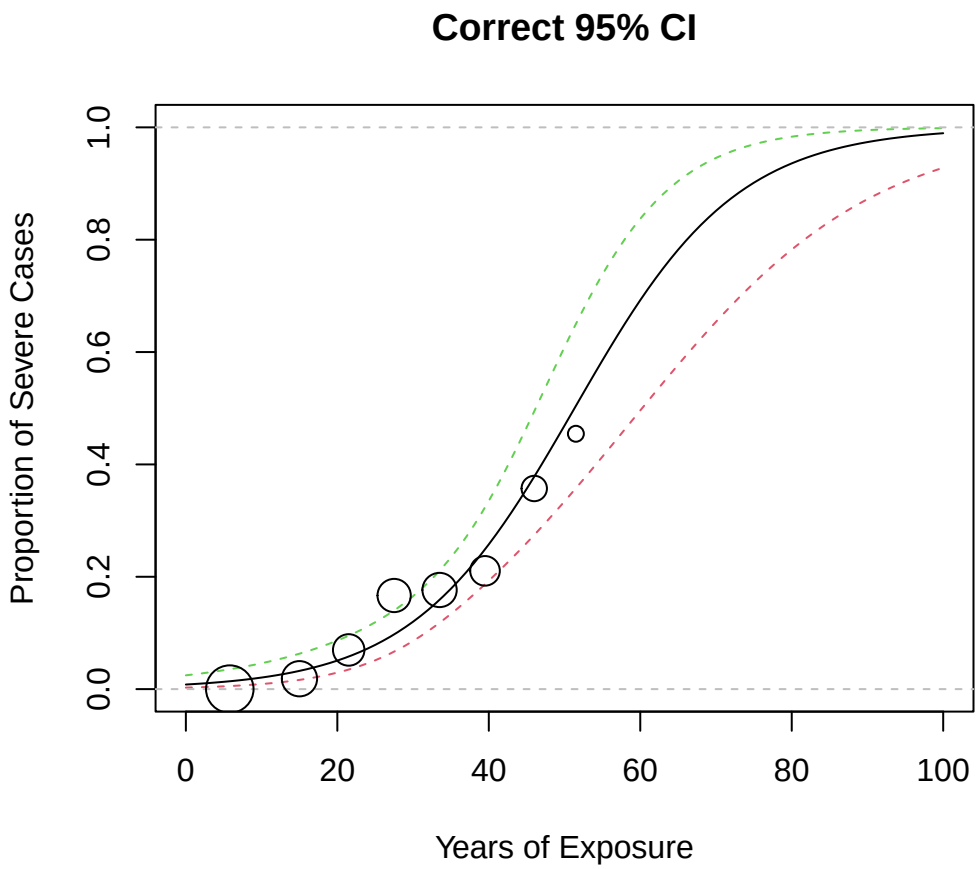


Correct 95% CI

```
pred <- predict(m4, newdata = pneu_new,
                type = "link", se.fit = TRUE)

pneu_ci <- cbind(fit = pred$fit,
                 lower = pred$fit - 1.96 * pred$se.fit,
                 upper = pred$fit + 1.96 * pred$se.fit) |> m4$family$linkinv() # plogis()

matplot(pneu_new$year, pneu_ci, type = 'l', lty = c(1, 2, 2), ylim = c(0, 1),
        xlab = "Years of Exposure", ylab = "Proportion of Severe Cases",
        main = "Correct 95% CI")
abline(h = 0:1, lty = 2, col = "gray")
points(case/n ~ year, pneu_gp, cex = sqrt(n)/3)
```



> **Note**
>
> GLM 的預測值在 linear predictor 的尺度下會更接近常態分布，因此 logistic regression 應該在 log odds 的尺度下計算信賴區間，之後再利用 link function 的反函數將 log odds 的信賴區間轉換為機率的信賴區間。

2.4 Comparison with linear model

```
library(ggplot2)

ggplot(pneu_gp, aes(x = year, y = case/n, weight = n)) +
  geom_hline(yintercept = 0:1, alpha = 0.6, linetype = 2) +
  geom_smooth(aes(colour = "GLM", fill = "GLM"),
              method = "glm", fullrange = TRUE,
```

```
            method.args = list(family = "binomial")) +
geom_smooth(aes(colour = "Linear", fill = "Linear"),
            method = "lm", fullrange = TRUE) +
geom_point(aes(size = n), alpha = 0.5) +
expand_limits(x = c(0, 100)) +
scale_colour_discrete(name = "Method") +
scale_fill_discrete(name = "Method") +
scale_size_binned() +
labs(x = "Years of Exposure", y = "Proportion of Severe Cases") +
theme_bw()
```

